NeuralSound

Ultrasound Simulation with Neural Operators

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Overview

- **Objective:** Develop an interactive simulator for Ultrasound imaging systems, modeling acoustic propagation with fixed latency.
- **Process:**
  - Input: Steering angle θ + TX pulse.
  - Process: Wave propagates through 2D heterogeneous tissue.
  - Output: 64-channel RF RX data.
  - Student is tasked to develop ideal TX Pulse and steering angle and is in charge of the image generation pipeline.
- **Challenge:**
  - Target Latency: $\sim$ 100 ms.
  - k-Wave (SOTA): 3–4 minutes (CFL stability limit).

Modeling the 2D Heterogeneous Medium

Acoustic propagation is governed by the mapping from medium parameters $\mathcal{A}$ to pressure fields $p(x, z, t)$.

- **Medium Maps:** $c(x, z)$ (Speed), $\rho(x, z)$ (Density), $M(x, z)$ (Inclusions).
- **Acoustic Impedance:** $Z(x, z) = \rho(x, z)c(x, z)$.
- **Governing Equation (Nonlinear):**

$$p = c_0^2 \left(\rho + \frac{B}{2A} \frac{\rho^2}{\rho_0} \right) - L\{\mathbf{u}, p\}$$

- Spatial variations in Z create the speckle and scattering required for imaging.

LTI Spatial Interpolation

- **Assumption:** In static tissue with limited nonlinear behavior, ultrasound propagation is approximately LTI.
- **Offline:**
  - Precompute spatial impulse responses with k-Wave
  - Sample over steering angle and medium state
  - Store responses in a lookup table
- **Online:**
  - Query nearby entries for a new angle or small tissue shift
  - Interpolate the stored response vectors in the frequency domain in real time
- **Benefit:** Near-perfect reconstruction fidelity with very low latency, around $5\,\mathrm{ms}$
- **Cost:** Memory scales poorly as the medium becomes more dynamic, heterogeneous, or nonlinear

LTI Results

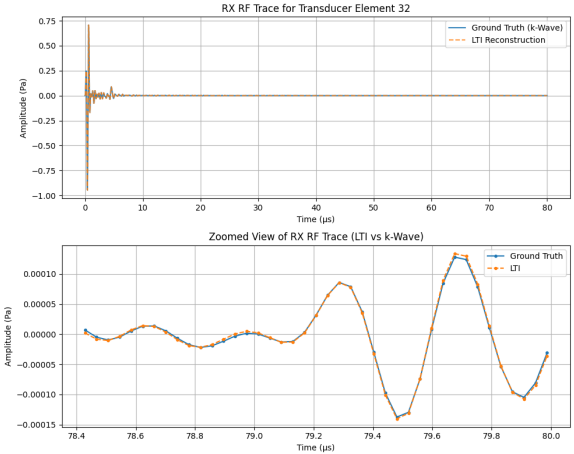


Figure: RF channel data: LTI vs. ground truth.

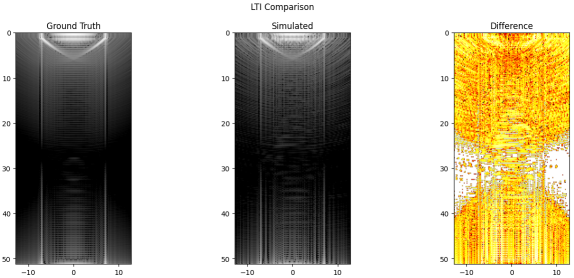


Figure: Beamformed image: low error.

Neural Operator + Architecture

- Attempt to approximate a nonlinear transfer function (i.e. function space to function space mapping using a FNO)

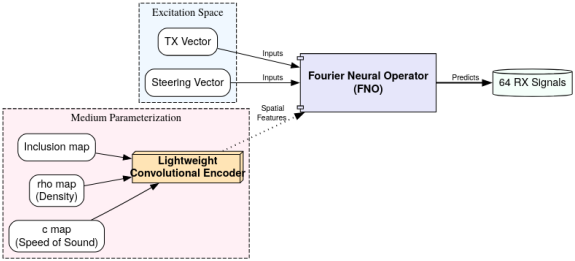


Figure: Architecture of the FNO used

Phase Issues

Standard reconstruction (MSE) is a poor metric for ultrasound quality.

- **Phase Sensitivity:** DAS beamforming relies on sub-wavelength alignment.
- **The Scale:** At 4 MHz, a 180° error is only $\sim$ 125 ns (or 0.2 mm).
- **Consequence:** Small phase shifts cause **destructive interference**, causing the B-mode image to collapse even if the signal "looks" correct in the time domain.

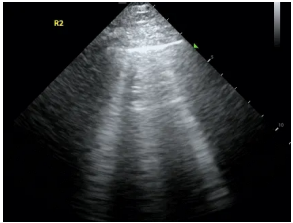
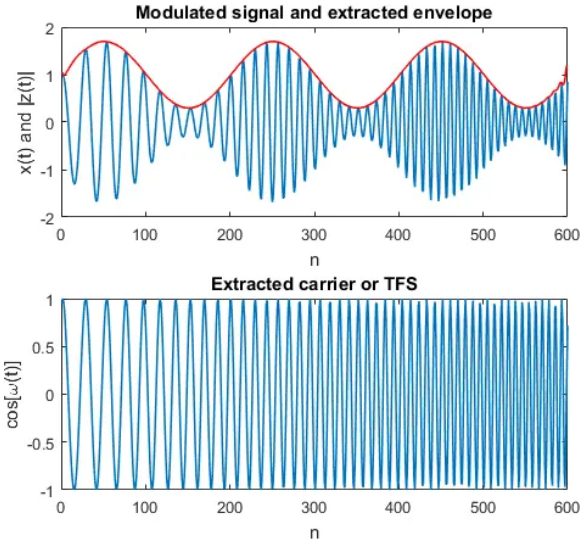


Figure: Lung Ultrasound Image Collapse

Fourier Neural Operator & IQ Modulation



- **Complexity Reduction:** Raw RF requires ~ 400 modes to resolve high frequencies.
- By using **IQ Modulation** ($s_{IQ} = I + jQ$), we shift the signal to a complex baseband.
- This reduces the computational requirement to just **16–32 modes**.

Designing for Coherence

To force the model to respect phase, we implemented an **Amplitude-Weighted Phase Loss**:

$$L_{wphase} = \frac{\sum_{\omega} |Y(\omega)| \, |\angle \hat{Y}(\omega) - \angle Y(\omega)|}{\sum_{\omega} |Y(\omega)|}$$

Total Training Objective

$$L_{total} = 0.8 L_{wphase} + 0.1 L_{spec} + 0.1 L_{mse}$$

Result: Significantly better stability than raw MSE, but FNO still struggled with localized transients.

FNO Results

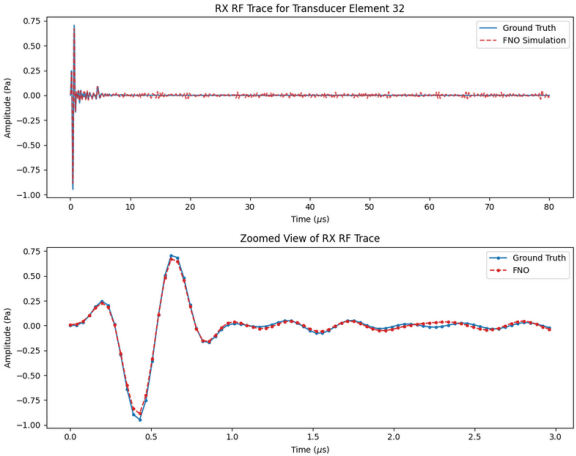


Figure: RF channel data: FNO vs. ground truth.

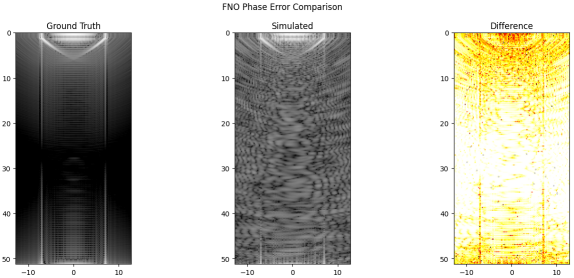


Figure: Beamformed image: phase errors cause destructive interference.

Wavelet Neural Operators

- **Wavelets:**
  - Fourier decompsition assumes stationary signals
  - Pulses are **transient** and **localized**
  - Stationary assumption fails and Fourier spaces are not the best appropriator of pulses
- **Multi-scale Decomposition:**

$$\mathcal{W}\{v\} = [c_A, c_D]$$

- c_A (Approximation): Models bulk wave travel and low-freq trends.
- c_D (Detail): Preserves sharp pulse edges and scattering interfaces.

WNO Results

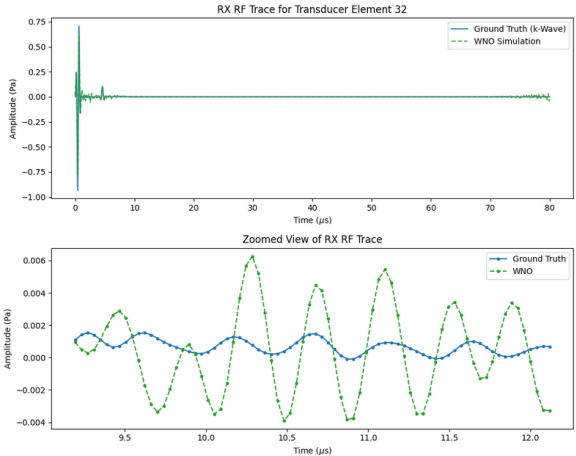


Figure: RF channel data: WNO vs. ground truth.

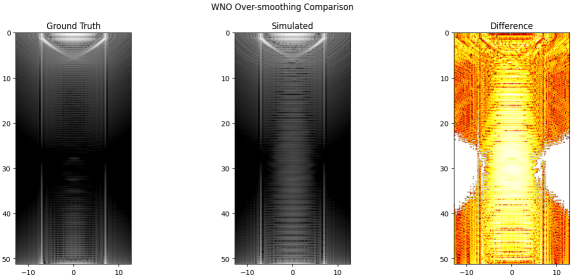


Figure: Beamformed image: pretty good reconstruction with some amplitude and smoothing differences.

Comparative Performance

Model	**Phase Error (Deg)**	**Latency**	**Image Quality**
k-Wave (GT)	0.0°	180s	Reference
FNO	$3.0 - 5.0^\circ$	<50ms	Artifact
WNO	$\mathbf{0.4}^\circ$	<100ms	Good
LTI Lookup	$<\mathbf{0.1}^\circ$	**5ms**	**Near Perfect**

- WNO drastically outperformed FNO in preserving phase coherence.
- In static media, the LTI interpolation outshines both NNs. Linear features dominate the system.

Conclusion

- **WNOs Capture Transient Pulses Better** Wavelets align with the physics of transient pulses better than Fourier modes for ultrasound.
- **MSE vs Phase Loss** Low MSE $\neq$ High Image Quality. Phase is the key value for beamforming and Ultrasound image quality
- **LTI Approximation Dominates** For static, linear regimes, classical DSP (LTI lookup & Frequency Domain Interpolation) provide the best speed and accuracy, at the cost of a large memory footprint.
- **Next Steps:**
  - Scaling WNO to 8k+ samples for complex inclusions.
  - C++/ONNX export for real-time integration into the C++ simulator.